

- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,
<https://womensmentalhealth.org/research/pregnancyregistry/atypical-antipsychotic>

Breast Feeding

- Unknown if lumateperone is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Recommended either to discontinue drug or bottle feed
- Infants of women who choose to breast feed while on lumateperone should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- For patients who do not tolerate other antipsychotics

- Patients requiring rapid onset of antipsychotic action without dosage titration
- Patients wishing to avoid drug-induced parkinsonism and akathisia
- Patients wishing to avoid weight gain and metabolic dysfunction such as dyslipidemia

Potential Disadvantages

- Dose adjustment not possible
- Expensive

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive symptoms
- Unstable mood (both depression and mania)
- Aggressive symptoms

Pearls

- Positive findings in two clinical trials in bipolar depression with approval pending for bipolar depression in patients with Bipolar I or II disorder as monotherapy and adjunctive therapy
- A phase 3 trial for treatment of agitation in patients with probable Alzheimer disease was discontinued after a pre-specified interim

analysis indicated that the study was not likely to meet its primary endpoint; the decision was not related to safety

- Impressions from clinical experience include observations of notable pro-social effects in many patients
- Pro-social effects include improved interpersonal relations; less isolation; more shopping, errands, cooking, walking; better organized in activities
- Neutral for weight gain in long-term studies and in clinical practice
- Favorable metabolic profile with changes in triglycerides, fasting glucose, and cholesterol similar to placebo in clinical trials and in clinical practice
- Neutral or placebo level of drug-induced parkinsonism and akathisia

Suggested Reading

Correll CU , Davis RE , Weingart M , et al. Efficacy and safety of lumateperone for treatment of schizophrenia: a randomized clinical trial . *JAMA Psychiatry* 2020 ;**77** :349–58 .
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Kumar B , Kuhad A , Kuhad A . Lumateperone: a new treatment approach for neuropsychiatric disorders . *Drugs Today (Barc)* 2018 ;**54** (12):713–19 .
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Lieberman JA , Davis RE , Correll CU et al. ITI-007 for the treatment of schizophrenia: a 4-week randomized, double-blind, controlled trial . *Biol*

Psychiatry 2016 ;**79** (12):952–61 .

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Snyder GL , Vanover KE , Zhu H . Functional profile of a novel modulator of serotonin, dopamine, and glutamate neurotransmission .

Psychopharmacology (Berl) 2015 ;**232** (3):605–21 .

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Vanover KE , Davis RE , Zhou Y . Dopamine D2 receptor occupancy of lumateperone (ITI-007): a positron emission tomography study in patients with schizophrenia . *Neuropsychopharmacology* 2019 ;**44** (3):598 –605 .

Lurasidone

Therapeutics

Brands

- Latuda
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: dopamine, serotonin receptor antagonist (DS-RAn)
- Atypical antipsychotic (serotonin-dopamine antagonist; second-generation antipsychotic; also a potential mood stabilizer)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia (ages 13 and older)**
- **Bipolar depression (monotherapy, ages 10 and older)**
- **Bipolar depression (adjunct, adults)**
- Acute mania/mixed mania

- Other psychotic disorders
- Bipolar maintenance
- Treatment-resistant depression
- Behavioral disturbances in dementia
- Behavioral disturbances in children and adolescents
- Disorders associated with problems with impulse control
- Posttraumatic stress disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis and stabilizing affective symptoms
- Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognition and affective symptoms
- Interactions at a myriad of other neurotransmitter receptors may contribute to lurasidone's efficacy
- Potently blocks serotonin 7 receptors, which may be beneficial for mood, sleep, cognitive impairment, and negative symptoms in schizophrenia, and also in bipolar disorder and major depressive disorder
- Partial agonist at 5HT1A receptors, and antagonist actions at serotonin 7 and alpha 2A and alpha 2C receptors, which may be beneficial for mood, anxiety, and cognition in a number of disorders

- Lacks potent actions at dopamine D1, muscarinic M1, and histamine H1 receptors, theoretically suggesting less propensity for inducing cognitive impairment, weight gain, or sedation compared to other agents with these properties

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition
- For psychosis, classically recommended to wait at least 4–6 weeks to determine efficacy of drug, but in practice some patients require up to 16–20 weeks to show a good response, especially on negative or cognitive symptoms
- For bipolar depression, onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks

If It Works –for Schizophrenia

- Most often reduces positive symptoms but does not eliminate them
- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
- Most schizophrenia patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenia patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year

- Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis
- For second and subsequent episodes of psychosis, treatment may need to be indefinite
- Even for first episodes of psychosis, it may be preferable to continue treatment indefinitely to avoid subsequent episodes

If It Works –for Bipolar Depression

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission) or significantly reduced
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite

If It Doesn’t Work –for Schizophrenia

- Try one of the other atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- Some patients may require treatment with a conventional antipsychotic
- If no first-line atypical antipsychotic is effective, consider higher doses or augmentation with valproate or lamotrigine
- Consider noncompliance and switch to another antipsychotic with fewer side effects or to an antipsychotic that can be given by depot injection
- Consider initiating rehabilitation and psychotherapy
- Consider presence of concomitant drug abuse

If It Doesn't Work –for Bipolar Depression

- Some patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)

Best Augmenting Combos for Partial Response or Treatment Resistance

- Valproic acid (valproate, divalproex, divalproex ER)
- Mood-stabilizing anticonvulsants (see [drug interactions](#))
- Topiramate
- Lithium
- Benzodiazepines

Tests

Before starting any atypical antipsychotic

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient
 - is overweight (BMI 25.0–29.9)
 - is obese (BMI >30)
 - has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
 - has diabetes (fasting plasma glucose >126 mg/dL)
 - has hypertension (BP >140/90 mmHg)

- has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting any atypical antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight
- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic
- * Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness, and clouding of sensorium, even coma

- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and lurasidone should be discontinued at the first sign of decline in WBC in the absence of other causative factors (class warning)
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

- Dose-dependent sedation

- Akathisia
- Nausea
- Dose-dependent hyperprolactinemia
- May increase risk for diabetes and dyslipidemia
- Tardive dyskinesia (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration



Life-Threatening or Dangerous Side Effects

- Tachycardia, first-degree AV block
- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics (class warning)
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain



- Many experience about one- to two-pound weight gain greater than placebo in short-term 6-week trials
- Patients in long-term 52-week trials actually lost 1.5 pounds on average
- Clinical experience, however, is still limited
- Appears to be less weight gain than observed with some antipsychotics
- Many patients lost weight in long-term trials when switching from olanzapine to lurasidone

Sedation



- May be higher in short-term trials than in long-term use

What to Do About Side Effects

- Wait
- Wait
- Wait
- Anticholinergics may reduce drug-induced parkinsonism when present
- Dose reduction may reduce akathisia when present

- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Consider changing to nighttime dosing (with evening meal)
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 40–80 mg/day for schizophrenia
- Some patients with schizophrenia may benefit from doses up to 160 mg/day

- 20–60 mg/day for bipolar depression
- Some patients with bipolar depression may benefit from doses up to 120 mg/day

Dosage Forms

- Tablet 20 mg, 40 mg, 60 mg, 80 mg, 120 mg

How to Dose

- Initial 40–80 mg once daily with food for schizophrenia
- Dose titration to initial dose of 40 mg/day is not required for schizophrenia
- Consider dose increases from 40 mg/day up to 160 mg/day as necessary and as tolerated for schizophrenia
- Initial 20 mg once daily with food for bipolar depression
- Can titrate to 120 mg/day in bipolar depression if necessary and tolerated



Dosing Tips

- Lurasidone should be taken with food (i.e., at least a small meal of a minimum of 350 calories)
- Lurasidone absorption can be decreased by up to 50% on an empty stomach and more consistent efficacy will be seen if dosing is done regularly with food

- Once daily dosing
- Giving lurasidone at bedtime can greatly reduce daytime sedation, akathisia, and drug-induced parkinsonism
- Starting dose for schizophrenia is 40 mg/day, which may be an adequate dose for early-onset psychosis cases
- 40–80 mg per day was suggested by controlled clinical trials as adequate for many patients with schizophrenia
- Some patients with schizophrenia benefit from higher dosing, with controlled clinical trials up to 160 mg/day
- Higher doses than 160 mg/day may benefit more difficult patients with schizophrenia with treatment nonresponsiveness to other agents
- Higher dosing, however, may cause more side effects
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Dosing for bipolar depression or major depressive episodes with mixed features is generally lower than for schizophrenia, with a 20 mg starting dose; 20 mg/day may be an adequate dose for some patients

- Doses between 20 and 60 mg/day are generally as efficacious as doses between 60 and 120 mg/day for bipolar depression or major depressive episodes with mixed features
- Some patients with bipolar depression or major depressive episodes with mixed features may require higher doses
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below 1000/mm³

Overdose

- Limited data

Long-Term Use

- Not extensively studied past 52 weeks, but long-term maintenance treatment is often necessary for schizophrenia
- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Down-titration, especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid discontinuation could theoretically lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Half-life 18–31 hours (shorter half-life better documented at the 40 mg dose)
- Metabolized by CYP450 3A4
- Cmax and bioavailability are reduced if taken without food

Drug Interactions

- Inhibitors of CYP450 3A4 (e.g., nefazodone, fluvoxamine, fluoxetine, ketoconazole) may increase plasma levels of lurasidone
- Coadministration of lurasidone with a strong CYP450 3A4 inhibitor (e.g., ketoconazole) or with a strong CYP450 3A4 inducer (e.g., rifampin) is contraindicated
- Coadministration of lurasidone with moderate CYP450 3A4 inhibitors can be considered; recommended starting dose is 20 mg/day; recommended maximum dose is 80 mg/day
- Moderate inducers of CYP450 3A4 may decrease plasma levels of lurasidone
- May enhance the effects of antihypertensive drugs

- May antagonize levodopa, dopamine agonists

Other Warnings/Precautions

- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and lurasidone should be used cautiously in patients at risk for aspiration pneumonia
- Monitor patients for activation of suicidal ideation, especially children and adolescents
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is taking a strong CYP 3A4 inhibitor (e.g., ketoconazole) or inducer (e.g., rifampin)
- In patients with a history of angioedema
- If there is a proven allergy to lurasidone

Special Populations

Renal Impairment

- Moderate and severe impairment: initial 20 mg/day; maximum dose 80 mg/day

Hepatic Impairment

- Moderate impairment: initial 20 mg/day; maximum dose 80 mg/day
- Severe impairment: initial 20 mg/day; maximum dose 40 mg/day

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension, although low potency at alpha 1 receptors suggests this risk may be less than for some other antipsychotics
- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist
- Lurasidone does not have a warning for QTc prolongation

Elderly

- In general, no dose adjustment is necessary for elderly patients
- However, some elderly patients may tolerate lower doses better
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events
- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic



Children and Adolescents

- Approved for schizophrenia in adolescents ages 13 and older
- Although lurasidone can be initiated at 40 mg/day, often a good idea to initiate at 20 mg to test for tolerability before raising dose to 40 mg, especially in children and in small body weight adolescents
- Approved for bipolar depression in bipolar I disorder in adolescents ages 10 and older
- Little or no weight gain documented in long-term studies up to 2 years
- Children and adolescents using lurasidone may need to be monitored more often than adults and may tolerate lower doses

better

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Animal studies do not show adverse effects; not teratogenic in rats and rabbits at doses up to 1.5–6 times the maximum recommended human dose
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Lurasidone may be preferable to anticonvulsant mood stabilizers if treatment is required during pregnancy
- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,

<https://womensmentalhealth.org/research/pregnancyregistry/atypicalantipsychotic>

Breast Feeding

- Unknown if lurasidone is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed unless the potential benefit to the mother justifies the potential risk to the child
- Infants of women who choose to breast feed while on lurasidone should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- Patients requiring rapid onset of antipsychotic action without dosage titration
- Patients who wish to take an antipsychotic once a day
- Patients experiencing weight gain from other antipsychotics or who wish to avoid weight gain
- Children with schizophrenia or bipolar disorder, as excellent tolerability in long-term studies documented

Potential Disadvantages

- Patients who cannot take a medication consistently with food

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive symptoms
- Unstable mood (both depression and mania)
- Aggressive symptoms

Pearls

- Clinical trials suggest that lurasidone is well tolerated with a favorable balance of efficacy and safety
- One of the few “metabolically friendly” antipsychotics
 - Neutral for weight gain (1–2 pounds weight gain in short-term studies, with 1–2 pounds weight loss in long-term studies)
 - Neutral for lipids (triglycerides and cholesterol)
 - Neutral for glucose
- Only atypical antipsychotic documented not to cause QTc prolongation, and one of the few atypical antipsychotics without a QTc warning
- Seems to have low-level drug-induced parkinsonism, especially when dosed at bedtime
- Somnolence and akathisia are the most common side effects in short-term clinical trials of schizophrenia that dosed lurasidone in